The question

MarinFold predicts contacts from a single sequence. The regime where that has something to prove is the one where MSA-based methods run out of signal: proteins with few homologs.

PR #257 produced our best decontaminated checkpoint (#232 `m2-p06` training, step 363,000) and scored it on `eval-val`, `eval-denovo`, and the legacy 554. This experiment finishes the read — `eval-test`, previously unscored for this checkpoint — and then bins every natural protein in the eval universe by the depth of the ColabFold MSA an MSA-based competitor would have had.

What ran

887 evaluation units on twelve single-H100 CoreWeave shards: the legacy 554 plus all 333 scorable FoldBench monomers. 100 rollouts per unit under the fixed #82 recipe, 88,700 usable, zero unfinished, 9m41s.

Depth comes off the two Modal volumes of ColabFold a3m files that Protenix's `+MSA` arm already ran with, measured through #74's `msa_depth.py` — raw sequence count and Neff, one code path for both volumes.

Validation

PR #257's published legacy-554, `eval-val`, and `eval-denovo` aggregates are the gate: same weights, same worker, so a disagreement would mean the path moved, not the model. All twelve reproduced, largest difference 0.0044.

The depth measurements match #247's independent count on all 314 FoldBench natural proteins exactly. Of the 11 stems both Modal volumes hold, 9 agree exactly and all 11 land in the same tier, so the three-week gap between the two ColabFold runs does not move the stratification.

The eval sets

subset	n	R (all)	R (long)
eval-test (first read for this checkpoint)	217	**0.5693**	0.5464
eval-val	97	0.5561	0.5402
eval-denovo	19	0.6110	0.5745
legacy 554	554	0.6059	0.5566

+0.032 on eval-test over the #232 sweep checkpoint, and 0.144 clear of the seq-KNN null over its own decontaminated corpus. Protenix-v2 + MSA scores 0.8446 on the same 217 proteins.

First: 15 "natural" proteins are designs

The CAMEO-hard / CASP-FM half was labelled natural by provenance, not by checking. RCSB says 15 of those 58 are de novo designs — and 13 of the 15 sit under MSA depth 10, because a designed protein has no homologs by construction.

They had to come out. Designed backbones are easy for structure predictors: in the shallow bin Protenix-v2 *single-sequence* scores 0.722 on the 13 designs against 0.241 on the 16 natural proteins. Pooling them produced a materially wrong conclusion in the first cut of this experiment — the same failure #241 found in eval2-natural. The natural universe is 357, not 372.

MSA depth — the corrected picture

Mean all-range R-precision, 357 natural proteins:

depth	n	MarinFold	Protenix-v2 + MSA	ESMFold2	Protenix-v2 single-seq
<10	16	0.300	0.336	0.426	0.241
10–100	32	0.279	0.755	0.469	0.273
100–1000	76	0.413	0.819	0.671	0.278
≥1000	233	0.616	0.858	0.827	0.249

MarinFold degrades with depth like everything else — depth proxies how well a family is represented anywhere, AFDB included.

What survives — the gap closes where the MSA goes

Paired per-protein against Protenix-v2 + MSA, 95 % bootstrap:

depth	n	MarinFold − (+MSA)
<10	16	**−0.036 [−0.180, +0.104]**
10–100	32	−0.477 [−0.545, −0.397]
100–1000	76	−0.407 [−0.454, −0.360]
≥1000	233	−0.242 [−0.263, −0.220]

At depth under 10 MarinFold is statistically level with the MSA-based model. The gap closes mostly because Protenix falls (0.858 → 0.336), which is what the single-sequence thesis predicts. It never leads: ESMFold2 is ahead in every tier, including this one (−0.126 paired).

Supporting checks: Protenix-v2 + MSA collapses toward its own single-sequence arm in this bin (0.336 vs 0.241), which is what an empty alignment should do; and the depths reproduce #247's independent count exactly on all 314 FoldBench naturals.

The low-MSA-depth cut — now a standing report

29 proteins under depth 10, with a `designed` column: 16 natural (11 CAMEO/CASP + 5 FoldBench) and 13 CAMEO designs. Frozen as `data/low_msa_depth_set.csv`, required by the `eval-checkpoint` skill, and browsable case by case in `dashboard/index.html`.

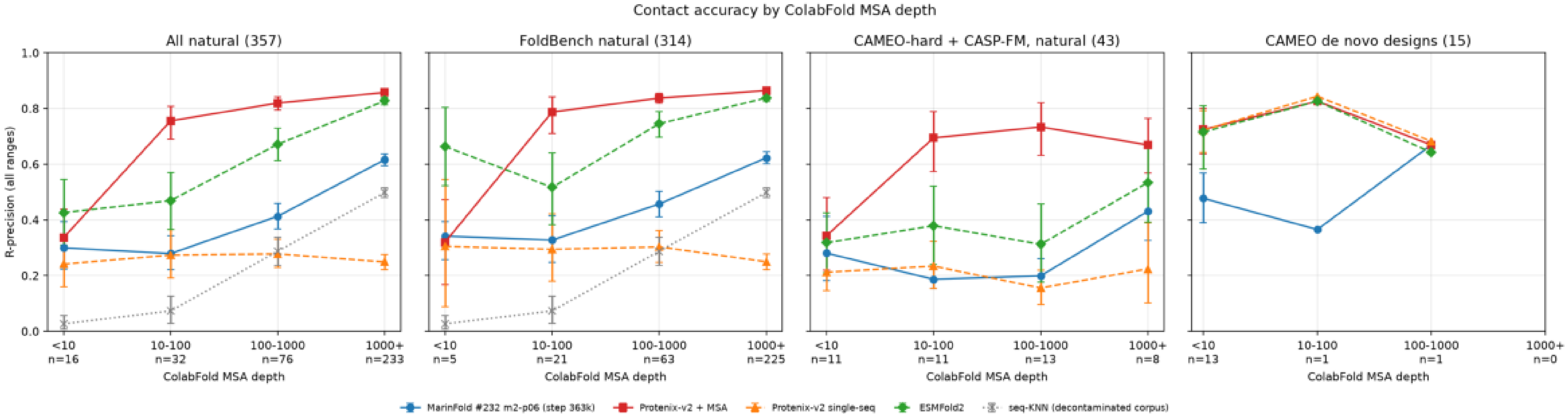
The CAMEO and CASP targets are generally inside the baselines' training sets, so the only like-for-like comparison in this regime is the 5 FoldBench members. Growing that is the follow-up this experiment argues for.

Caveats

- 16 natural proteins under depth 10, only 5 of them uncontaminated for the baselines, is too thin to carry the headline on its own. - Median length is 148 residues in the <10 bin against 290 at ≥1000, and contact prevalence goes as ~1/L. That bias favours the shallow bins, so the reported decline is conservative.

rprecision_by_depth_tier.png

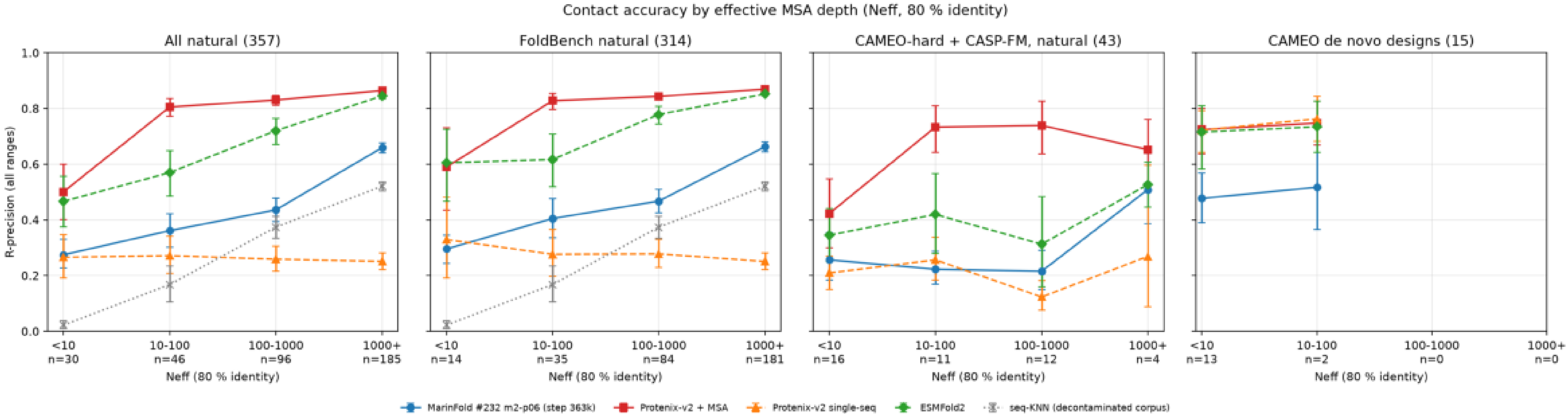
Mean all-range R-precision per MSA-depth tier, 95 % bootstrap intervals; de novo designs are held out of the natural panels.



$ plot_depth.py

rprecision_by_neff_tier.png

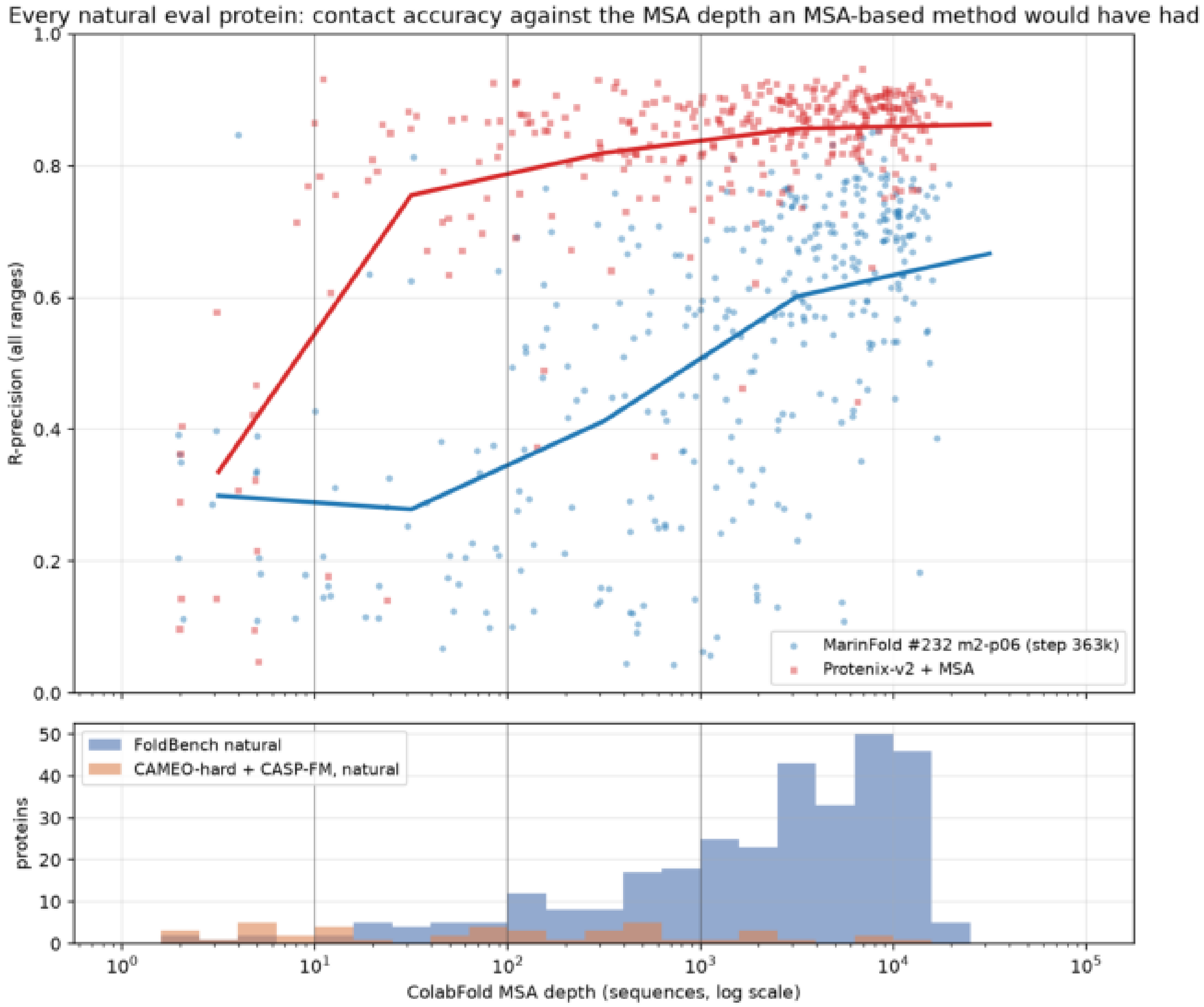
The same cut on redundancy-weighted depth, which fills the shallow bins raw sequence count leaves nearly empty.



$ plot_depth.py

rprecision_vs_depth_scatter.png

Every natural eval protein, accuracy against depth, with per-decade means; the histogram shows where each subset sits.



$ plot_depth.py